GIRTH-FIVE MATCHING-PREFIX CERTIFICATES IN THE SEARCH FOR A DEGREE-57 MOORE GRAPH

ARGUS OPEN RESEARCH PROJECT

Abstract. By the Hoffman–Singleton theorem, a Moore graph of diameter 2 and girth 5 can exist only for degrees $k \in \{2, 3, 7, 57\}$, with $k = 57$ on $v = k^2 + 1 = 3{,}250$ vertices remaining open. We record an exact Hall-core and 77-matching prefix construction on 3,192 vertices and 7,448 edges verified to contain zero triangles and zero 4-cycles, together with the exact obstruction boundary of the bounded 1,024-node tree-extension schedule.

1. Verified Partial Prefix Properties

Proposition 1.1. *In the rooted tree decomposition of a hypothetical* 57*-regular Moore graph on* 3,250 *vertices from a base edge, the reconstructed* 77*-matching prefix subgraph on* 3,192 *vertices and* 7,448 *edges has girth at least* 5 *(zero* 3*-cycles and zero* 4*-cycles), while the fixed* 1,024*-node Hall-core extension branch terminates without completing all* 56×56 *inter-branch matchings.*

Date: August 29, 2026.
2020 *Mathematics Subject Classification.* Primary 05C35; Secondary 05C70, 05E30.